

Yunyi Li

Ph.D. Candidate
McCombs School of Business
The University of Texas at Austin
Cell: (319) 519-3209
Email: Yunyi.Li@mcombs.utexas.edu
Web: yunyli.github.io

Education

- **Ph.D. in Information, Risk and Operations Management** Expected: May 2026
The University of Texas at Austin
 - Thesis: Trustworthy AI and Data Quality: Understanding and Mitigating Label Errors
 - Advisors: Maytal Saar-Tsechansky and Maria De-Arteaga
- **M.S. in Informatics-Information Science** May 2020
The University of Iowa
- **B.B.A in Business Analytics with Honors (top 1%)** May 2018
The University of Iowa
 - Large Data Analysis Certificate
 - Study Abroad: National Taiwan University of Science and Technology, Taiwan, 2015

Research Interests

Trustworthy AI, Data Integrity, Label Bias, Interpretable Machine Learning, Human-AI Collaboration, AI for Social Good, Human-Centered Design

Selected Publications

1. Wang, T., Yang, J., Li, Y., and Wang, B., *PIE—Partially Interpretable Estimators with Refinement*, **Informations Journal of Computing (IJOC)**, 2025
2. Li, Y., De-Arteaga, M., and Saar-Tsechansky, M., *Label Bias: A Pervasive and Invisibilized Problem*, **Notices of American Mathematical Society**, 2024
3. Srivastava, S., Xu, Z., Li, Y., Street, W. N., and Gilbertson-White, S., *Gaussian process regression and classification using International Classification of Disease codes as covariates*, **Stat**, 2023
4. Li, Y., De-Arteaga, M., and Saar-Tsechansky, M., *More Data Can Lead Us Astray: Active Data Acquisition in the Presence of Label Bias*, **AAAI Conference on Human Computation and Crowdsourcing (HCOMP)**, 2022
5. Gilbertson-White, S., Srivastava, S., Li, Y., Laures, E., Saeidzadeh, S., Yeung, C., and Chae, S., *Multimorbidity and cancer: using electronic health record (EHR) data to cluster patients in multimorbidity phenotypes*, **Journal of Pain and Symptom Management (JPSM)**, 2019
6. Wang, Y., Wang, A., Liu, Z., Thurman, A., Powers, L., Zou, M., Hefel, A., Li, Y., Zabner, J., and Au, K.F., *Single-molecule long-read sequencing reveals the chromatin basis of gene expression*, **Genome Research**, 2019
7. Li, Y., and Wang, T., *Next Hit Predictor - Self-exciting Risk Modeling for Predicting Next Locations of Serial Crimes*, **NeurIPS AI for Social Good**, 2018

Paper Under Review

Li, Y., De-Arteaga, M., and Saar-Tsechansky, M., *Bias-Aware Mislabeling Detection via Decoupled Confident Learning*. Preparing for **Major Revision at Information Systems Research**.

- 2025 Conference on Information Systems and Technology (CIST)
- Invited talks at the 21st Big XII MIS Research Symposium and the McCombs School of Business Research Spotlight
- 2024 INFORMS Annual Meeting
- 2024 Symposium on Statistical Challenges in Electronic Commerce, Research (SCECR)
- 2024 Good Systems Symposium
- 2023 International Conference on Machine Learning (ICML)

Work in Progress

1. Li, Y., Gao R. and Saar-Tsechansky, M., *Human-LLM Collaboration for Scalable Data Annotation*
2. Li, Y., De-Arteaga, M., and Jarvenpaa, S., *Trust at the Edge — Clarify Uncertainty of AI systems*

Selected Awards

- **McCombs School of Business Dean's Fellowship** - The University of Texas at Austin (2020–2025)
- **UT Austin Machine Learning Laboratory Research Grant** - The University of Texas at Austin (2021)
- **Good Systems Award for Ethical AI Research** - The University of Texas at Austin (2022)
- **The University of Iowa Graduate College Research Fund** - The University of Iowa (2019)
- **Big Data Grant from National Science Foundation (NSF)** - The University of Iowa (2017–2018)
- **Carol Fethke Honors Scholarship** - The University of Iowa (2018)
- **Iowa Center for Research by Undergraduates Research Fellowship** - The University of Iowa (2017)
- **Dean's List** - The University of Iowa (2015–2018)

Selected Talks

1. **Conference on Information Systems and Technology (CIST)**, October 2025
2. **INFORMS Information Systems Society Doctoral Consortium**, October 2025
3. **21st Big XII MIS Research Symposium (Invited Talk)**, April 2025
4. **2024 INFORMS Annual Meeting**, October 2024
5. **Symposium on Statistical Challenges in Electronic Commerce Research (SCECR)**, June 2024
6. **2024 Good Systems Symposium**, March 2024
7. **McCombs School of Business Research Spotlight (Invited Talk)**, November 2023
8. **2023 INFORMS Annual Meeting**, October 2023

9. International Conference on Machine Learning (ICML), July 2023
10. AAAI Conference on Human Computation and Crowdsourcing, November 2022
11. Annual Conference on Neural Information Processing Systems (NeurIPS), December 2018

Professional Services

- **Conference Session Chair:** INFORMS Annual Meeting 2024
- **Peer reviewer:**
 - International Conference on Information Systems (ICIS) 2023
 - Hawaii International Conference on System Sciences (HICSS) 2022 & 2023
- **Conference Assistant:** International World Wide Web Conference (WWW) 2023

Research Experiences

- **Graduate Researcher** The University of Texas at Austin, 08/2020–Present
McCombs School of Business
Topic: Trustworthy AI and Data Quality – Understanding and Mitigating Label Errors
 - Led and published the research paper *More Data Can Lead Us Astray: Active Data Acquisition in the Presence of Label Bias*
 - Invited, first-authored, and published the article *Label Bias: A Pervasive and Invisibilized Problem*, shedding light on the often-overlooked role of biased labeling in data-driven systems
 - Proposed a novel and principled machine learning framework- Decoupled Confident Learning (DeCoLe), for bias-aware mislabeling detection
- **Research Assistant** The University of Iowa, 12/2018–05/2020
Department of Statistics
Topic: Statistical Learning of Electronic Health Records (EHR) to Cluster Cancer Patients by Multimorbidity Profiles
 - Identified common clusters of chronic conditions co-occurring in patients with advanced solid tumor cancer and described differences across these clusters
 - Designed a kernel-based distance function for cancer patient data using electronic health records
 - Employed advanced statistical learning methods, including Gaussian Processes and Gibbs Sampling, to solve regression and classification problems involving cancer patient data
- **Research Intern** The University of Iowa, 05/2018–09/2018
Carver College of Medicine
Topic: Single-Molecule Long-Read Sequencing Reveals the Chromatin Basis of Gene Expression
 - Created visualizations using R for chromatin accessibility and nucleosome positioning
 - Generated an Rcircos landscape plot to display multiple features at the whole-genome scale
 - Applied a Hidden Markov Model and dynamic programming to analyze signal distribution

- **Research Project Lead**

The University of Iowa, 12/2016–07/2018

Tippie College of Business

Topic: Next Hit Predictor – Self-Exciting Risk Modeling for Predicting Serial Crime Locations

- Designed a novel spatio-temporal crime prediction model for the Cambridge, MA Police Department and formulated a convex learning objective based on pairwise rankings and trained the model using stochastic gradient descent (SGD)
- Presented research at NeurIPS 2018, Iowa Research Open House, Data Mining Iowa Group, and the Iowa Dare to Discover Campaign Showcase

Teaching Experiences

***Completed Teaching Training and Supervised Teaching at McCombs School of Business**

1. **Artificial Intelligence & Machine Learning for Executives: MIS 281N**, Teaching Assistant
The University of Texas at Austin (Spring 2025)
2. **Intro to Information Technology Management: MIS 301**, Teaching Assistant
The University of Texas at Austin (Spring 2025)
3. **Ethics of AI: BAX 372**, Teaching Assistant and Course Co-designer
The University of Texas at Austin (Spring 2024)
4. **Intro to Problem Solving and Programming: MIS 304**, Teaching Assistant
The University of Texas at Austin (Fall 2020 & Fall 2022)
5. **Data Science: MSCI 6070**, Teaching Assistant
The University of Iowa (Spring 2020)
6. **Advanced Database Management & Big Data: MSCI 4220**, Teaching Assistant
The University of Iowa (Spring 2020)
7. **Data Management and Visual Analytics: MSCI 6050**, Teaching Assistant
The University of Iowa (Fall 2018)
8. **Big Data Summer School**, Student Mentor
The University of Iowa (Summer 2017 & Summer 2018)

Leadership Experiences

- **McCombs Ph.D. Council**

McCombs School of Business, 09/2022–07/2023

- Assisted in planning and organizing the McCombs School of Business Centennial Celebration
- Represented the interests and concerns of Ph.D. students to the administration to ensure their voices were heard

- **Graduate Student Assembly**

The University of Texas at Austin, 09/2021–07/2022

- Participated in university-wide policy setting for the Graduate School and reviewed academic programs
- Served as a voting member in Graduate School decision-making processes

- **2nd Place, USITCC Business Analytics Competition** San Antonio, TX, 04/2018
U.S. Information Technology Collegiate Conference (USITCC)

 - National undergraduate information technology competition held annually
 - Competed in two events: awarded 2nd place in Business Analytics and received Honorable Mention in Database Management
- **Team Leader, MinneMUDAC Competition** The University of Iowa, 08/2017–05/2018
Annual Midwest Undergraduate Data Analytics Competition (MinneAnalytics) — Eden Prairie, MN

 - Led a team of four business students to develop data-driven solutions for real-world problems
 - Applied machine learning techniques to analyze the relationship between water quality and housing prices